

project: KztekAdbPublishTool (multi-project solution)

last_updated: 2026-08-10

updated_by: Senior Developer

CODE-GRAPH — KztekAdbPublishTool Solution

Lịch sử cập nhật

Ngày	Người	Thay đổi
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2026-08-05	Senior Developer	Tạo mới — Phase 1 Foundation (STEP 1.1–1.3)
2026-08-10	Senior Developer	Cập nhật đầy đủ sau Phase 2+3 hoàn tất; thêm <code>DeviceState.Remove()</code> (bug fix R9); bổ sung toàn bộ module Services/Endpoints/State còn thiếu

1. Tổng quan solution

Project	SDK	TargetFramework	Vai trò
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KztekAdbPublishTool	Sdk.WinForms	net8.0-windows7.0	Bản gốc WinForms — KHÔNG SỬA
KztekAdbPublishTool.Web	Sdk.Web	net8.0	Bản migrate mới — ASP.NET Core Razor Pages
KztekAdbPublishTool.Web.Tests	Sdk	net8.0	Unit tests cho project Web

2. Project: KztekAdbPublishTool.Web

2.1 Cấu trúc thư mục

```
src/KztekAdbPublishTool.Web/  
├─ Configuration/  
│   └─ AdbSettings.cs          ← POCO config (AdbPath, PollIntervalMs, DbPath,  
UploadsPath, MaxUploadBytes)  
├─ Endpoints/  
│   ├── ApkEndpoints.cs        ← POST /api/apk/upload, DELETE /api/apk  
│   └── DeviceEndpoints.cs     ← POST /api/devices/{connect,connect-  
batch,remove,poll}, GET /api/devices, POST /api/settings/package, /api/polling/toggle
```

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├── HealthEndpoints.cs      ← GET /health
├── InstallEndpoints.cs     ← POST /api/install
├── ScanEndpoints.cs        ← POST /api/scan/start, /api/scan/cancel
├── Hubs/
│   └── DeviceHub.cs        ← SignalR Hub, endpoint /hubs/device
├── Models/
│   └── DeviceRecord.cs     ← POCO entity (9 property), dùng làm DB row +
SignalR DTO
├── Pages/
│   └── Index.cshtml        ← Dashboard: toolbar 3 hàng, grid 9 cột, action
panel, log area
│   └── Index.cshtml.cs     ← IndexModel : PageModel
│       └── Shared/
│           └── _Layout.cshtml ← Bootstrap 5 local, navbar KZTEK brand (Navy/Cam)
├── Services/
│   └── AdbService.cs        ← Singleton; wrap adb binary;
GetDevices/Connect/Install/LaunchApp/GetVersion
├── ApkManifestReader.cs    ← Singleton; parse AXML từ .apk; stateless
├── DeviceRepository.cs     ← Singleton; SQLite CRUD
(Upser/Remove/GetAll/GetSetting/SetSetting/UpdateInstallResult)
├── InstallCoordinator.cs   ← Singleton; queue install per-device
(SemaphoreSlim(1)), global (SemaphoreSlim(4))
├── PollControlService.cs   ← Singleton; PollingEnabled flag + TriggerAsync
(manual poll trigger)
├── ScanCoordinator.cs      ← Singleton; TCP probe scan (SemaphoreSlim(24), max
512 IP, timeout 1200ms)
├── ScanRangeParser.cs      ← Static; parse IP range string → List<string> IPs
├── State/
│   └── DeviceState.cs      ← Singleton; ConcurrentDictionary in-memory
snapshot; AddOrUpdate/TryGet/GetAll/GetSerials/Remove
├── Workers/
│   └── DevicePollWorker.cs  ← BackgroundService; poll 3s → Upsert →
RefreshVersion → push SignalR DevicesUpdated
├── wwwroot/
│   └── js/
│       ├── dashboard.js    ← Main UI logic: SignalR handlers, device grid,
install/remove/connect buttons
│       ├── scan-modal.js   ← Network scan modal JS
│       └── signalr-client.js ← SignalR client (auto-reconnect, expose
window.kzHubConnection)
│           └── lib/
│               ├── bootstrap.bundle.min.js ← Bootstrap 5 local (tránh CDN fail trong Docker
LAN)
│               └── signalr.min.js          ← SignalR client lib local
├── Program.cs              ← Entry point, DI registration, Kestrel+FormOptions
500MB limit
├── appsettings.json
└── appsettings.Development.json

```

2.2 Module Dependencies

Module	Phụ thuộc vào	Được gọi bởi (Callers)	Confidence	Last verified
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Program.cs	AdbSettings, AdbService, DeviceRepository,	— (entry point)	CONFIRMED	2026-08-10

	ApkManifestReader, DevicePollWorker, DeviceHub, DeviceState, InstallCoordinator, ScanCoordinator, PollControlService			
Configuration/AdbSettings	—	Program.cs, AdbService, DeviceRepository, DevicePollWorker	CONFIRMED	2026-08-10
Services/AdbService	IOptions\<AdbSettings\>, System.Diagnostics.Process	DevicePollWorker, InstallCoordinator	CONFIRMED	2026-08-10
Services/DeviceRepository	IOptions\<AdbSettings\>, Microsoft.Data.Sqlite, Models/DeviceRecord	DevicePollWorker, DeviceEndpoints, InstallEndpoints, ApkEndpoints, InstallCoordinator	CONFIRMED	2026-08-10
Services/ApkManifestReader	System.IO.Compression	ApkEndpoints	CONFIRMED	2026-08-10
Services/InstallCoordinator	AdbService, DeviceRepository, DeviceState, IHubContext\<DeviceHub\>	InstallEndpoints	CONFIRMED	2026-08-10
Services/PollControlService	—	DevicePollWorker, DeviceEndpoints	CONFIRMED	2026-08-10
Services/ScanCoordinator	IHubContext\<DeviceHub\>, ScanRangeParser	ScanEndpoints	CONFIRMED	2026-08-10
Services/ScanRangeParser	—	ScanCoordinator	CONFIRMED	2026-08-10
State/DeviceState	System.Collections.Concurrent, Models/DeviceRecord	DevicePollWorker, DeviceEndpoints, InstallEndpoints, InstallCoordinator	CONFIRMED	2026-08-10
Hubs/DeviceHub	Microsoft.AspNetCore.SignalR.Hub	Program.cs (MapHub), DevicePollWorker, InstallCoordinator, ScanCoordinator	CONFIRMED	2026-08-10
Workers/DevicePollWorker	IOptions\<AdbSettings\>, AdbService, DeviceRepository, DeviceState, PollControlService, IHubContext\<DeviceHub\>	Program.cs (AddHostedService)	CONFIRMED	2026-08-10
Models/DeviceRecord	—	DeviceRepository, DeviceState, DevicePollWorker, InstallCoordinator, API JSON response	CONFIRMED	2026-08-10
Endpoints/DeviceEndpoints	DeviceRepository, DeviceState, AdbService, PollControlService	Program.cs (MapDeviceEndpoints)	CONFIRMED	2026-08-10
Endpoints/InstallEndpoints	DeviceState, DeviceRepository, InstallCoordinator	Program.cs (MapInstallEndpoints)	CONFIRMED	2026-08-10
Endpoints/ScanEndpoints	ScanCoordinator	Program.cs (MapScanEndpoints)	CONFIRMED	2026-08-10

Endpoints/ApkEndpoints	IOptions\<AdbSettings\>, ApkManifestReader, DeviceRepository	Program.cs (MapApkEndpoints)	CONFIRMED	2026-08-10
Endpoints/HealthEndpoints	—	Program.cs (MapHealthEndpoints)	CONFIRMED	2026-08-10
Pages/Index	IndexModel : PageModel	Router (/)	CONFIRMED	2026-08-10
wwwroot/js/signalr-client.js	signalr.min.js (local lib)	_Layout.cshtml	CONFIRMED	2026-08-10
wwwroot/js/dashboard.js	signalr-client.js, bootstrap.bundle.min.js (local)	_Layout.cshtml	CONFIRMED	2026-08-10
wwwroot/js/scan-modal.js	signalr-client.js	_Layout.cshtml	CONFIRMED	2026-08-10

2.3 API Endpoints

Method	Path	Handler	Status
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GET	/	Pages/Index	✓
GET	/health	HealthEndpoints	✓
WS/GET	/hubs/device/negotiate	DeviceHub (SignalR)	✓
POST	/api/install	InstallEndpoints → InstallCoordinator	✓
POST	/api/scan/start	ScanEndpoints → ScanCoordinator	✓
POST	/api/scan/cancel	ScanEndpoints → ScanCoordinator	✓
POST	/api/apk/upload	ApkEndpoints	✓
DELETE	/api/apk	ApkEndpoints	✓
POST	/api/devices/connect	DeviceEndpoints → AdbService + PollControlService	✓
POST	/api/devices/connect-batch	DeviceEndpoints → AdbService + PollControlService	✓
POST	/api/devices/remove	DeviceEndpoints → DeviceState + DeviceRepository	✓ Bug R9 fixed 2026-08-10
GET	/api/devices	DeviceEndpoints → DeviceRepository	✓
POST	/api/settings/package	DeviceEndpoints → DeviceRepository	✓
POST	/api/polling/toggle	DeviceEndpoints → PollControlService + DeviceRepository	✓
POST	/api/devices/poll	DeviceEndpoints → PollControlService.TriggerAsync	✓

2.4 SignalR Events (server → client)

Event	Payload	Triggered by	Status
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DevicesUpdated	List<DeviceRecord>	DevicePollWorker (mỗi 3s)	✓
InstallProgress	string serial, int percent, string message	InstallCoordinator	✓
DeviceInstalled	string serial, bool success, string version	InstallCoordinator	✓
ScanProgress	int found, int scanned, int total	ScanCoordinator	✓
ScanFound	string ipPort	ScanCoordinator	✓
ScanCompleted	int total, int found	ScanCoordinator	✓

2.5 Configuration (appsettings.json — section "Adb")

Key	Default (production)	Default (development)	Ghi chú
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AdbPath	/opt/platform-tools/adb	C:\platform-tools\adb.exe	Path tới adb binary
PollIntervalMs	3000	3000	Chu kỳ poll (ms)
DbPath	/app/data/adbpublishstool.db	adbpublishstool-dev.db	SQLite DB
UploadsPath	/app/uploads	uploads-dev	Thư mục APK tạm
MaxUploadBytes	500000000	500000000	500 MB
Kestrel MaxRequestBodySize	500000000	inherited	Set trong Program.cs

3. Project: KztekAdbPublishTool (WinForms nguồn — KHÔNG SỬA)

Không cập nhật graph của project này — bất khả xâm phạm theo §1A WF-MIGRATE.

4. Thay đổi gần đây

Ngày	File	Loại thay đổi
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2026-08-05	src/KztekAdbPublishTool.Web/**	Tạo mới toàn bộ project (Phase 1-3)
2026-08-05	tests/KztekAdbPublishTool.Web.Tests/**	Tạo mới test project + 5 test cases DeviceRepository
2026-08-05	KztekAdbPublishTool.sln	Thêm 2 project mới vào solution
2026-08-10	State/DeviceState.cs	Thêm method <code>Remove(string serial)</code> — fix bug R9: thiết bị không biến mất sau khi xóa
2026-08-10	Endpoints/DeviceEndpoints.cs	Handler <code>/api/devices/remove</code> nhận thêm <code>DeviceState</code> qua

DI, gọi
`deviceState.Remove()`
trước `repo.Remove()` —
parity WinForms
MainForm.cs:572-577